Analysis 1: Aug-Nov 2026

I am writing to let you all know that in the rest of the semester (21 oct when I will teach my next class and 20 nov, last day of classes), I am going to conduct at least 2, maybe 3, quizzes which will have a weight in the final grade. Earlier I was not planning it this way, but during the last 2 weeks before the mid-semester exam, I saw that the attendance was way below par. The Surprise quiz is a way to ensure better attendance.

Of course In case of medical issue if anyone has to miss the class (and hence the quiz) will find an alternative. Likewise for those who have taken prior permission to miss.

Here are some exercises you all should try to solve and discuss with each others and TAs.

Exercise

Prove all the assertions from first principles.
You can use results on convergence of sequences; continuity and uniform continuity of functions; root test, ratio test for power series, differentiability of functions, mean value theorem (but you must give precise statement(s)). Do not use any other standard result(s) without providing proof.

1. Let $g\colon \mathbb{R} \to \mathbb{R}$ be defined by $g(x) = |x^2 - 4| + |x - 2|$.
   (a) Determine all points $x \in \mathbb{R}$ where g fails to be differentiable.
   (b) Calculate the one-sided derivatives $g'_+(x_0) = \lim_{h\downarrow 0} \frac{g(x_0+h)-g(x_0)}{h}$ and $g'_-(x_0) == \lim_{h\uparrow 0} \frac{g(x_0+h)-g(x_0)}{h}$ at each non-differentiable point x_0.
2. Let $f\colon [a,b] \to \mathbb{R}$ be a function that is continuous on $[a,b]$ and twice differentiable on (a,b). Suppose f has three distinct roots in $[a,b]$, i.e., $f(x_1) = f(x_2) = f(x_3) = 0$ for $a \le x_1 < x_2 < x_3 \le b$.
   Prove that there exists at least one point $c \in (a,b)$ such that $f''(c) = 0$.
3. Let $f\colon [0,\infty) \to \mathbb{R}$ be a differentiable function such that $f(0) = 0$ and $f'(x)$ is strictly increasing for all $x > 0$.
   (a) Show that for all $x > 0$, $f(x) < xf'(x)$.
   (b) Define $g(x) = \frac{f(x)}{x}$ for $x > 0$. Prove that $g(x)$ is strictly increasing on $(0,\infty)$.
4. Let $n \ge 1$ be an integer and let f be a real-valued function defined on (a,b), and is n-times differentiable on (a,b). Suppose $f^{(n)}$ is continuous on (a,b). For $u, x \in (a,b)$ let

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(u)}{k!}(x-u)^k.$$

   and let $R_n(u) = f(u) - P_n(u)$ (P_n is called the Taylor Polynomial of f around u and R_n is the remainder). Show that
   (a) R_n can be expressed as $R_n(x) = H_n(x)(x-u)^n$ for a funtion H_n that satisfies $\lim_{x\to u} H_n(x) = 0$.
   (b) Suppose $f^{(n)}$ is differentiable on the interval I between u and x ($I = (u,x)$ if $u < x$ and $I = (x,u)$ if $x < u$). Then $\exists \xi \in I$ such that

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{k!}(x-u)^{(n+1)}$$